

Networks

Costs

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Optimization: principles and algorithms



Costs

Moving flow along an arc generate costs
In our formalism, we consider proportional costs.
Modeling costs is not necessarily easy.
We discuss two common issues in this video

Costs



Costs

$$c_{ij}, \forall (i, j) \in \mathcal{A}.$$

Data of the problem

- ▶ Cost of transporting one unit of flow on arc (i, j) .
- ▶ Unit is arbitrary.
- ▶ Total cost for arc (i, j) :

$$c_{ij}x_{ij}.$$

- ▶ Total cost for the network:

$$\sum_{(i,j) \in \mathcal{A}} c_{ij}x_{ij}.$$

Generalized cost



Generalized cost



Toll road

- ▶ travel time (say 30 minutes),
- ▶ travel cost (say 10 CHF).

Generalized cost



Toll road

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Value of time

30 CHF/hour or 0.5 CHF/min.

Generalized cost

- ▶ In CHF: $30 \text{ min} \times 0.5 \text{ CHF/min} + 10 \text{ CHF} = 25 \text{ CHF}$.
- ▶ In min: $30 \text{ min.} + 10 \text{ CHF} / (0.5 \text{ CHF/min}) = 50 \text{ min.}$

Link additivity

Cost of a path

$$C(P) = \sum_{(i,j) \in P^{\rightarrow}} c_{ij} x_{ij} - \sum_{(i,j) \in P^{\leftarrow}} c_{ij} x_{ij}.$$

Cost along a simple path flow

$$\begin{aligned} C(P) &= \sum_{(i,j) \in P^{\rightarrow}} f c_{ij} - \sum_{(i,j) \in P^{\leftarrow}} f c_{ij} \\ &= f \left(\sum_{(i,j) \in P^{\rightarrow}} c_{ij} - \sum_{(i,j) \in P^{\leftarrow}} c_{ij} \right). \end{aligned}$$

Example



GVA → ZRH → BKK

CHF 2412

GVA → ZRH

CHF 570

ZRH → BKK

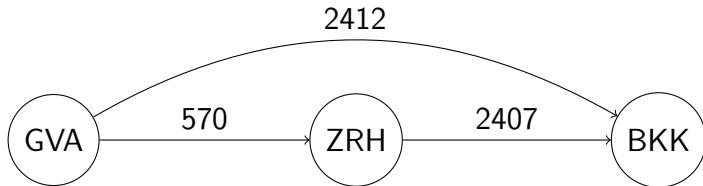
CHF 2407

Modeling

First model



Second model



Summary

Costs are important data of flow problems

In general, we want to optimize total costs

In our mathematical framework, costs must all be in the same unit

And they must be link additive.